Task 2: Continuous Variables

COURSE NAME

CSDS 413 Introduction to Data Analysis

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Contents

Context

In this exercise, our aim is to quantify the associations between continuous variables and assess the statistical significance of these associations. For this purpose, we will use the three datasets that are provided with the assignment:

- The file 'p2a.csv' contains a matrix of size 2400 x 2 which has 2400 samples and 2 variables.
- The file 'p2b.csv' contains a matrix of size 110 x 2 which has 110 samples and 2 variables.
- The file 'p2c.csv' contains a matrix of size 2100 x 2 which has 2100 samples and 2 variables.

1 Part (a)

For the two variables provided in "p2a.csv", assess the association between them by computing Pearson correlation r_a and computing a p-value p_a for the null hypothesis of no association. Select a significance level α and reject the null-hypothesis if the p-value is less than α. Explain, in complete sentences, your findings: Is there a statistically significant association (at α level) between the provided variables? What is the magnitude and the direction of the association?

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Below is the function and the tabulation for r_a and p_a:

```
def compute_pearson_correlation(x: np.ndarray, y: np.ndarray) -> tuple:
    """
    Computes Pearson correlation coefficient and p-value.

    :param x: First variable
    :type x: np.ndarray
    :param y: Second variable
    :type y: np.ndarray
    :returns: Tuple of (correlation coefficient, p-value)
    :rtype: tuple
    """
    r, p_value = pearsonr(x, y)
    return r, p_value
```

Figure 1: Function for computing Pearson correlation and p-value.

Pearson r	P-value
0.3809	1.04×10^{-83}

Figure 2: Pearson correlation results for p2a.

The Pearson correlation given by the function was 0.380875. Given a selected significance level $\alpha = 0.05$, there is a statistically significant, positive association between the two variables. The p value is incredibly low, magnitudes lower than the selected α, but 0.05 is convention and follows most of the class exercises we have done. By no means is this result challenged by that threshold, so we can confidently reject the null hypothesis.

2 Part (b)

Repeat part (a) for the variable pair provided in "p2b.csv" and compute Pearson correlation r_b and p-value p_b. Compare the Pearson correlations r_a and r_b as well as the p-values p_a and p_b. Explain your findings: Which variable pair (in part a or b) has a stronger association according to the comparison of the correlations? Which variable pair has a stronger association according to the comparison of the p-values? Do the comparisons according to correlation coefficients and p-values agree on which variable pair indicate the same stronger association? If not, why is there such a discrepancy? Next, draw scatter plots (variable 1 vs. variable 2) to visualize the data for both part (a) and part (b). Which variable pair (in part a or b) has a stronger association do you think according to the scatter plots? Does your conclusion agree with the comparisons of the p-values and correlation coefficients? If not, explain why.

Below is the tabulation of the Pearson correlations and p values as well as scatter plots for each dataset:

Dataset	Pearson r	P-value
p2a	0.3809	1.04×10^{-83}
p2b	0.9312	3.74×10^{-49}

Figure 3: Comparison of Pearson correlation results between p2a and p2b.

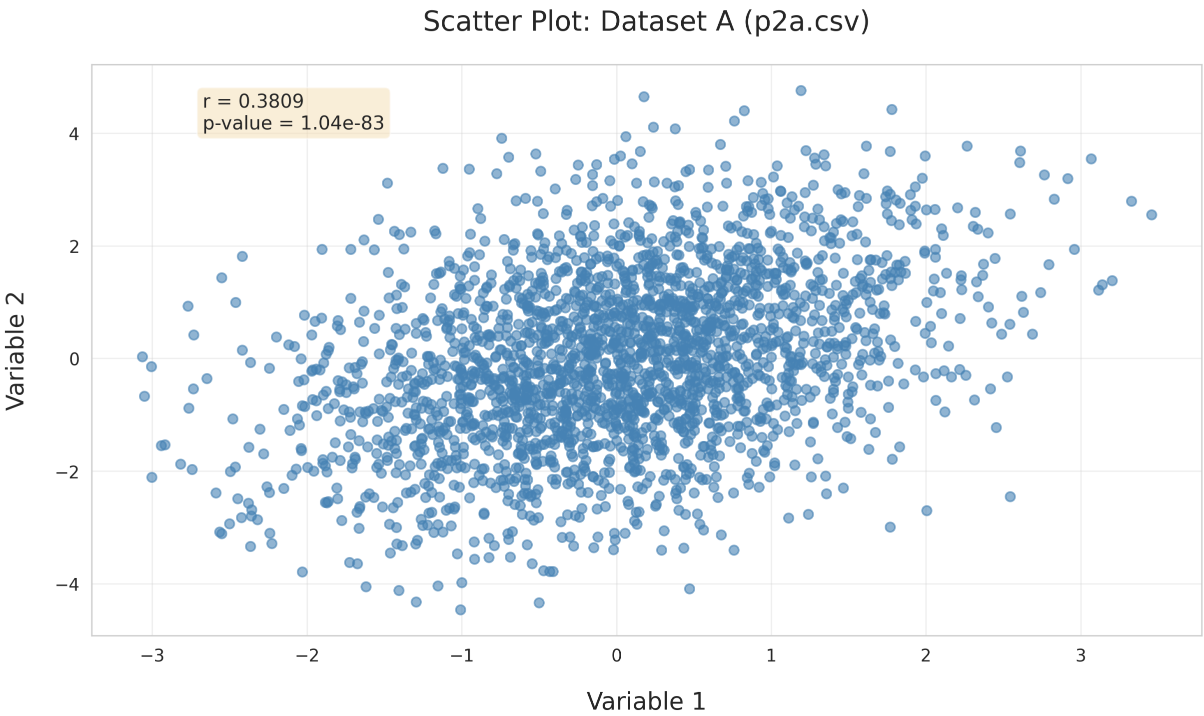

Figure 4: Scatter plot for p2a.csv.

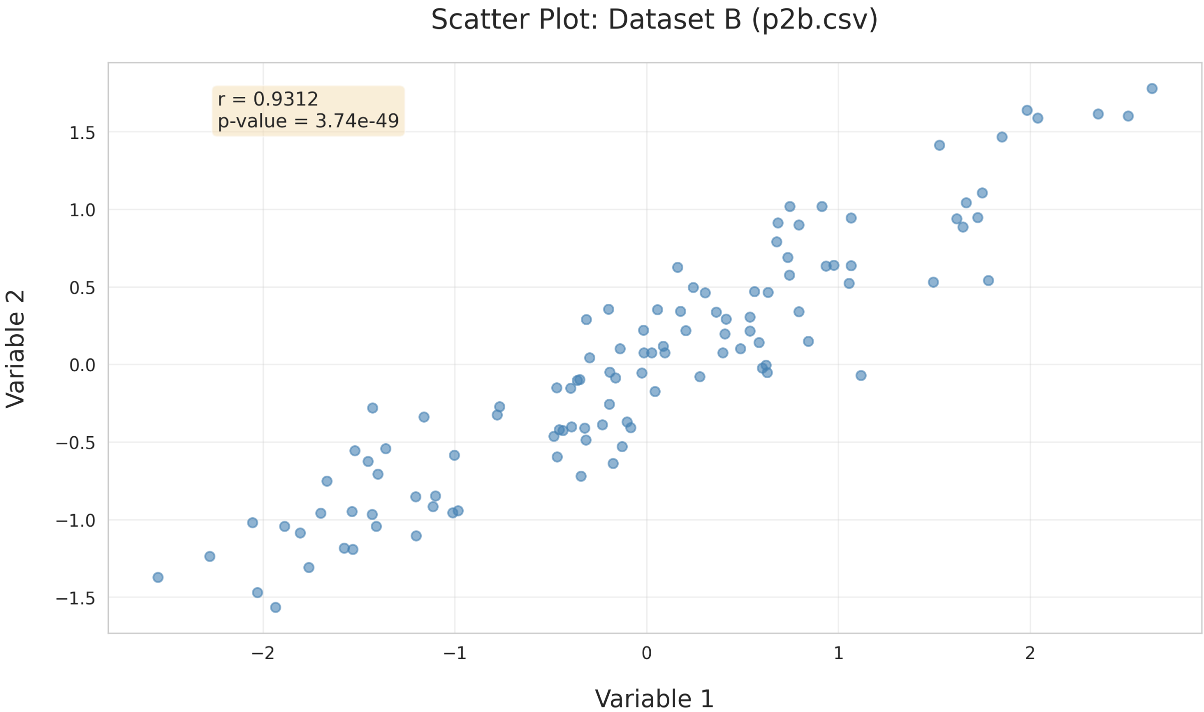

Figure 5: Scatter plot for p2b.csv.

The Pearson correlation for p2b came to 0.9312, and given the same significance level $\alpha = 0.05$, there is a statistically significant, positive association between the two variables. The p value $p_b = 3.74 \times 10^{-49}$ is again much lower than the threshold, so we can confidently reject the null hypothesis.

According to the raw correlations $r_a = 0.3809$ and $r_b = 0.9312$, the part b variable pair shows a stronger correlation. However, the p-values $p_a = 1.04 \times 10^{-83}$ and $p_b = 3.74 \times 10^{-49}$ suggest that the part a variable pair is much smaller and therefore shows a stronger association, so these criteria don't agree. The reason for this is the difference in statistical power that lies in the sample size. p2a contains 2400 samples, while p2b only has 110: so in the case of p2a, even a moderate correlation becomes much more statistically significant because there is much more evidence that the pattern in not occurring by chance, whereas p2b has much less data to rule out randomness at play contributing to the association, and so even such an extreme correlation cannot produce a lower p-value.

Regarding the scatter plots, I would say off visual inspection that p2b shows a much clearer relationship, albeit the graph is much more sparse. The data stick very closely to the trend line as you would expect for such a high correlation, whereas p2a shows some agreement generally in the positive direction but as a whole kind of bulges out and looks befitting of a sample that would yield a moderate correlation value. So, my visual assessment agrees with the correlation assessment and not the p-value assessment. The p-value in p2a is simply a product of a larger sample and ought not to be misconstrued as an indication of a stronger relationship. Not to mention that the strength of p-values does not lie in making claims about qualities of the data itself, but more so in the uncertainty we possess in such claims. For that reason, I think my visual assessment of the scatter plots makes sense.

3 Part (c)

Repeat part (b). But, this time compare the associations in the datasets "p2a.csv" and "p2c.csv". Make sure to answer the questions in part (b), this time for comparing the variable pairs in parts (a) and (c).

Below is the tabulation of the Pearson correlations and p values as well as scatter plots for each dataset:

Dataset	Pearson r	P-value
p2a	0.3809	1.04×10^{-83}
p2c	0.0412	0.0592

Figure 3: Comparison of Pearson correlation results between p2a and p2c.

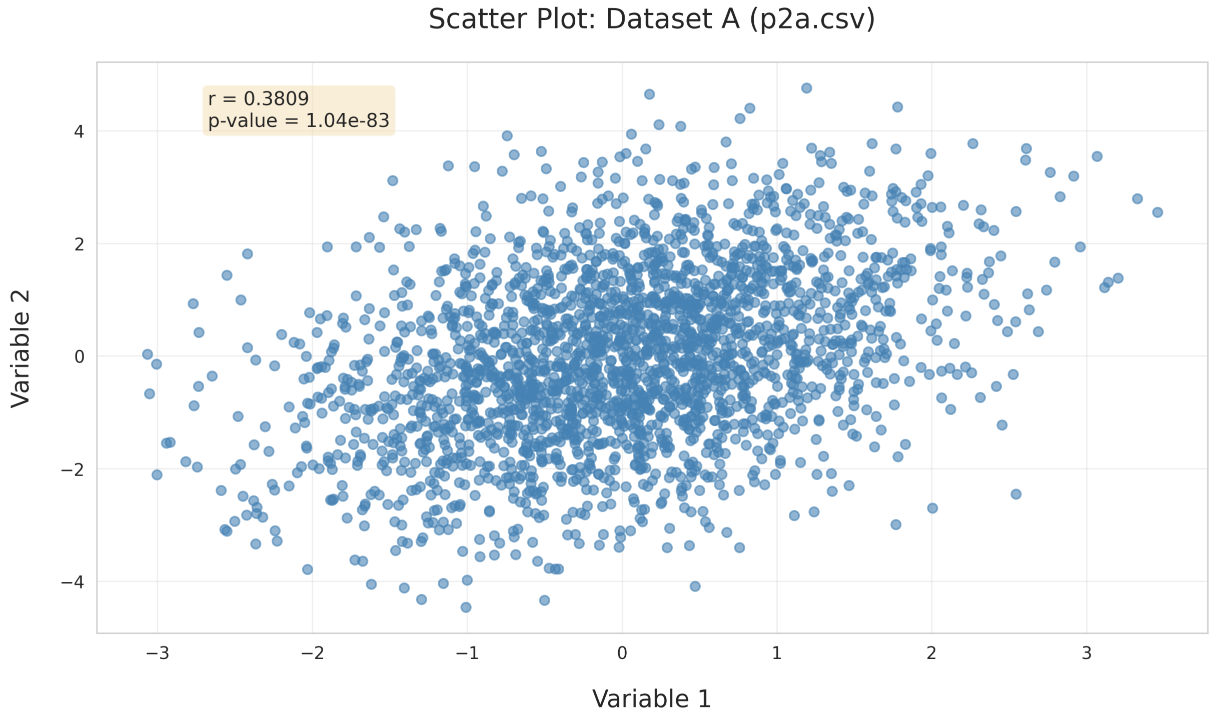

Figure 4: Scatter plot for p2a.csv.

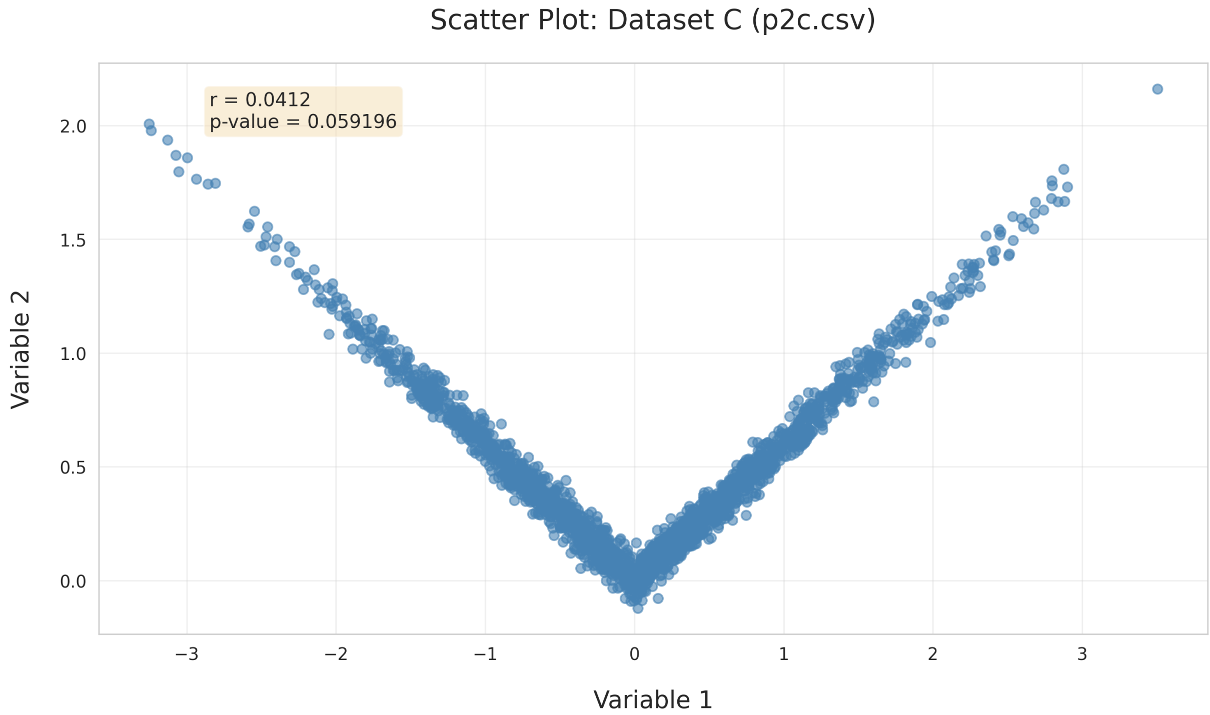

Figure 5: Scatter plot for p2c.csv.

The Pearson correlation for p2c came to 0.0412, and given the same significance level $\alpha = 0.05$, there is not a statistically significant association between the variables. Given the same significance level $\alpha = 0.05$, the p value $p_c = 0.0592$ is higher than the threshold, so we fail to reject the null hypothesis.

According to the raw correlations $r_a = 0.3809$ and $r_c = 0.0412$, the part a variable pair shows a stronger correlation. Similarly, the p-values $p_a = 1.04 \times 10^{-83}$ and $p_c = 0.0592$ suggest that the part a variable pair is much smaller and therefore shows a stronger association, so the criteria agree and there is no discrepancy.

As far as the scatter plots, again the p2a dataset shows a generally positive agreement that vaguely sticks to the trend line; a result you would expect from a moderate, positive Pearson correlation value. The p2c dataset however shows a very odd behavior: it seems as though as variable 1 increases and is negative, there is a very strong negative association conveyed by variable 2, and similarly for positive variable 1 values, variable 2 shows a very strong positive association. I would think that if p2c just possessed only negative variable 1 samples or only positive variable 1 samples, these results may be very different.

If we are to just go off of visual assessment, Pearson correlation aside and just considering the relationship between the variables, I would claim that the p2c variable pair shows much stronger association than p2a. This disagrees with both suggestions made by the Pearson correlations and the p-values. This exercise appears to be a demonstration of how Pearson correlation strictly quantifies linear associations and what scenarios it may fail in.

Interestingly enough, the occurrence of positive variable 1 samples and negative variable 1 samples is likely so even in p2c, the two relationships on each side of the y-axis almost perfectly cancel each other out, giving a correlation very close to zero, despite it being immediately apparent from the plot that a strong, non-linear association is present between the variables. That being said, the reason my visual assessment disagrees with the suggestions made by the Pearson correlations and p-values is that I can see from the scatter plot this non-linear relationship that Pearson correlation is blind to, and without the necessary nuance to identify it, the suggestions made by the Pearson correlations and the associated p-values are not compelling in this context.